Trend Following, Leveraged Re-Entry, and Volatility Decay

S&P 500 total return. One signal only: the 200-day moving average (the daily equivalent of Faber's 10-month rule). Tested as far back as the data allows.

Data. Total return throughout. Monthly S&P 500 total return back to 1901 (Shiller) for the Faber replication; a daily total-return series from 1928 for everything else — real `^SP500TR` from 1988, and before that `^GSPC` price plus the Shiller dividend yield (this reconstruction tracks the real series with 0.50%/yr error and 0.9996 correlation over 1988–2026). Cash earns the 13-week T-bill (`^IRX`; a 3.5% constant before 1960). The trend signal is lagged one day, so nothing uses information we could not have had.

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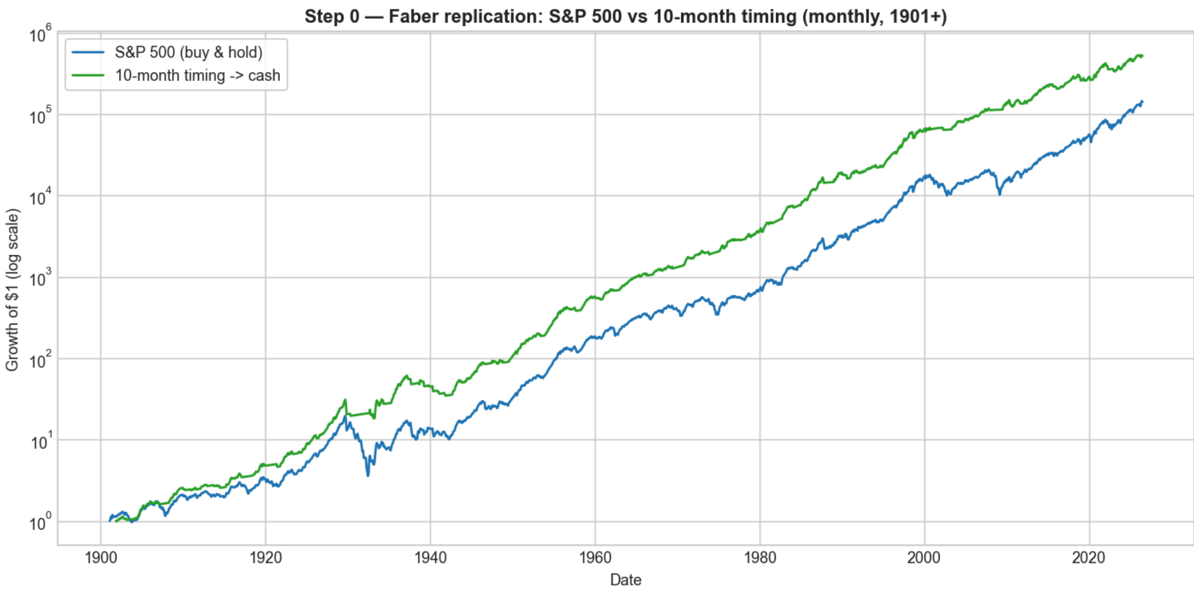
1. Buy & hold vs the Faber moving-average rule

The rule: hold the S&P 500 while it is **above** its moving average; move to **cash** while it is **below**. Faber uses the 10-month SMA on monthly data; we use that, then its daily twin (the 200-day SMA).

Monthly, 1901–2026 (Faber's exact setup):

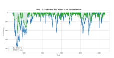
	S&P 500 buy & hold	10-month timing → cash
CAGR	9.95%	11.24%
Volatility	15.4%	10.8%
Sharpe	0.44	**0.69**
Max drawdown	–81.8%	**–43.0%**

(Faber's published drawdowns are –83.66% → –42.24%; we reproduce them to within a point.)



Daily, 200-day SMA, 1928–2026:

	Buy & Hold 1×	MA200 → Cash
CAGR	10.14%	11.29%
Volatility	18.9%	12.6%
Sharpe	0.40	**0.60**
Sortino	0.56	**0.84**
Max drawdown	–83.9%	**–46.2%**
Calmar	0.12	**0.24**

The moving-average rule keeps essentially all of the return while cutting volatility by a third and halving the worst drawdown, so its Sharpe, Sortino and Calmar are all much higher. The 200-day MA adds clear risk-adjusted value.

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2. Leverage returns

A daily **L× leveraged** return is simply that day's S&P 500 total return multiplied by L, and then compounded day by day:

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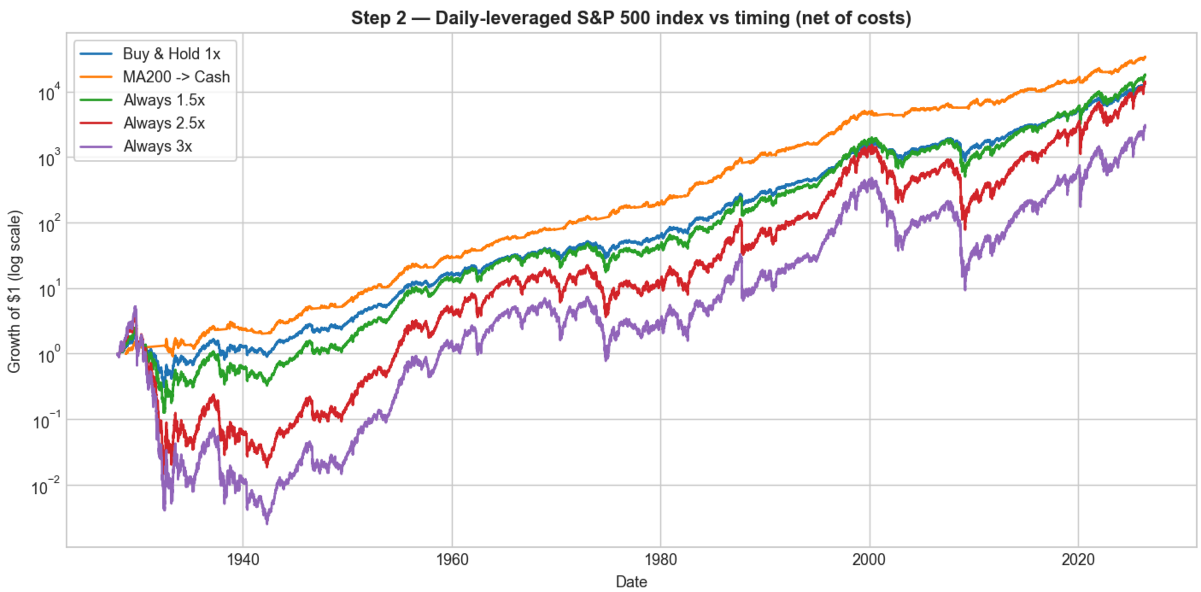
leveraged_return[t] = L × sp500_return[t]        (before fees / financing)
```

For example, the 2× series is just the daily S&P total return × 2, compounded. Borrowed money costs the financing rate, and leveraged funds charge a fee (~0.9%/ yr); both are included below.

Holding **constant** daily leverage on the index, net of costs:

	CAGR	Grew $1 to
1× (buy & hold)	10.14%	$13,021
Always 1.5×	10.46%	$17,378
Always 2.5×	10.15%	$13,110
Always 3×	**8.43%**	$2,803



Constant leverage barely helps at 1.5× and *loses ground* by 3× — across a full century (including 1929 and 2008), constant 3× ends below plain buy & hold and far below the moving-average rule.

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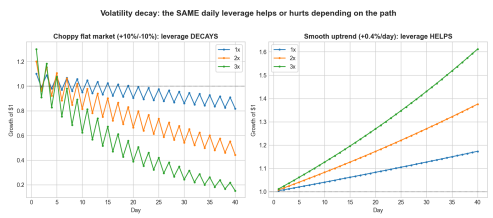
3. Volatility decay, and buying leverage at the lows

Volatility decay. A +10% day followed by a −10% day:

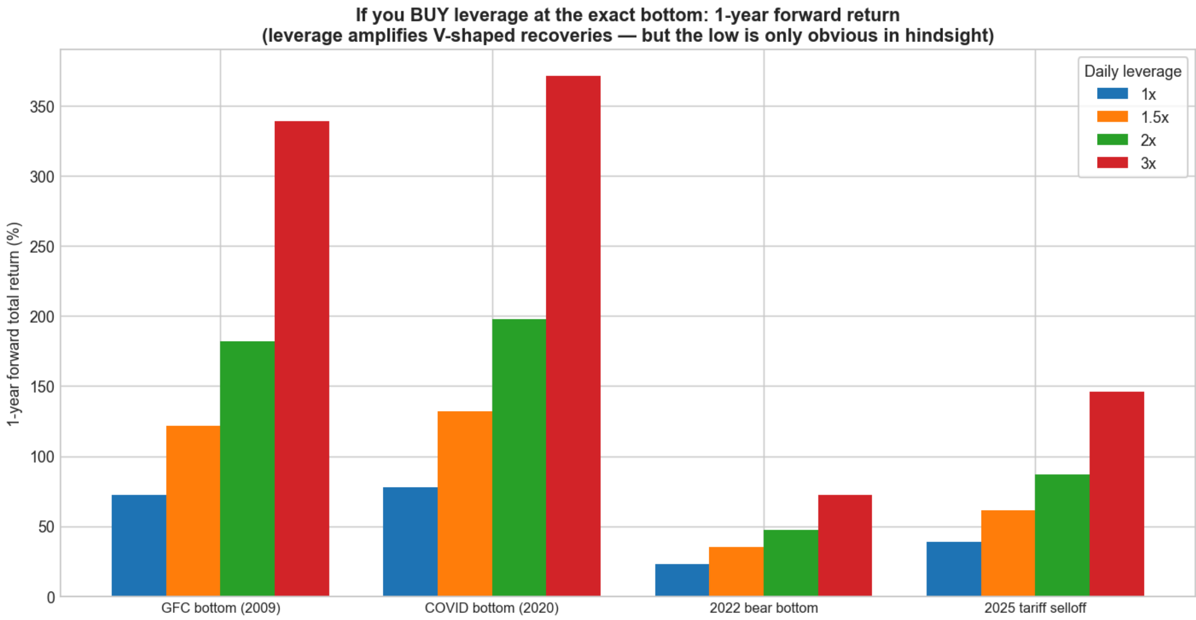
	1×	2×	3×
Two-day return	−1.0%	−4.0%	−9.0%

The market round-trips to roughly flat, but leverage loses — and the loss grows with the *square* of leverage. The annual penalty is ≈ $\frac{1}{2} \cdot L^2 \cdot \sigma^2$: at 20% volatility, ~2%/yr for 1×, **8%/yr for 2×, 18%/yr for 3×**.



But decay only bites in choppy/falling markets. In a one-directional rally — like the rebound off a market bottom — leverage amplifies the gain. Forward **1-year** total return if you had bought at the exact low:

Bottom	1×	1.5×	2×	3×
GFC (2009-03-09)	+72%	+122%	+182%	**+339%**
COVID (2020-03-23)	+78%	+132%	+198%	**+372%**
2022 (2022-10-12)	+23%	+35%	+47%	+72%
2025 tariff selloff (2025-04-08)	+39%	+61%	+87%	**+146%**



Leverage can work strongly in your favour **if you time it** — buying into the recovery off a low. (The low, of course, is only obvious in hindsight.)

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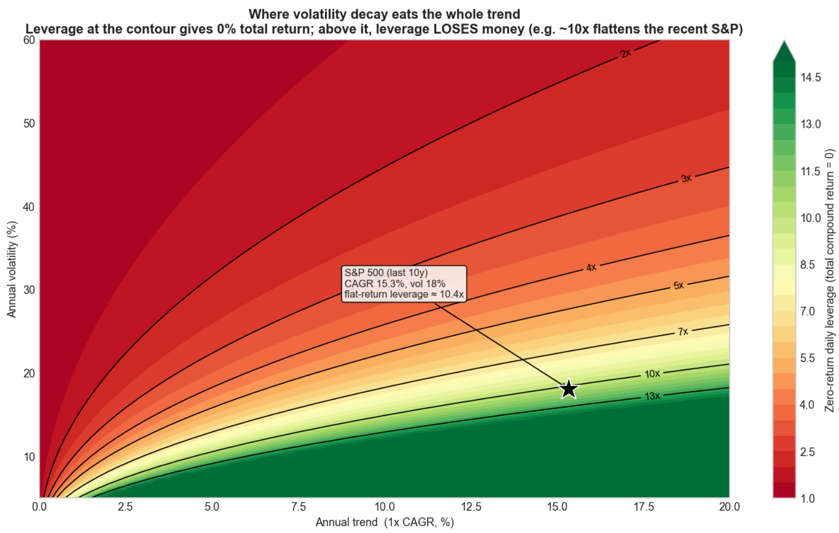
4. The leverage at which the total return goes flat

For a given trend (the 1× CAGR g) and volatility σ, the compound return of L× leverage is $L\cdot\mu - \frac{1}{2}\cdot L^2\cdot\sigma^2$. Setting it to zero gives the leverage at which volatility decay exactly cancels the trend, so the **total return is flat (0%)**:

$$L_{zero} = 2\cdot g / \sigma^2 + 1$$

Below it, leverage still grows; above it, leverage **loses money**. The map below plots `L_zero` for every combination of trend and volatility. For the S&P over the **last 10 years** (CAGR 15.3%, volatility 18.1%) the flat point is **≈ 10×** — so a hypothetical "10× S&P" would have gone essentially nowhere despite a strong decade, while higher still would have bled toward zero.



(For reference, the leverage that merely *ties* 1× buy & hold is lower — ≈ 3.1× for the S&P — and the growth-maximising level is ≈ 2×; see `charts/F3_breakeven_leverage_map.png` and `charts/F3_optimal_leverage_curve.png`.)

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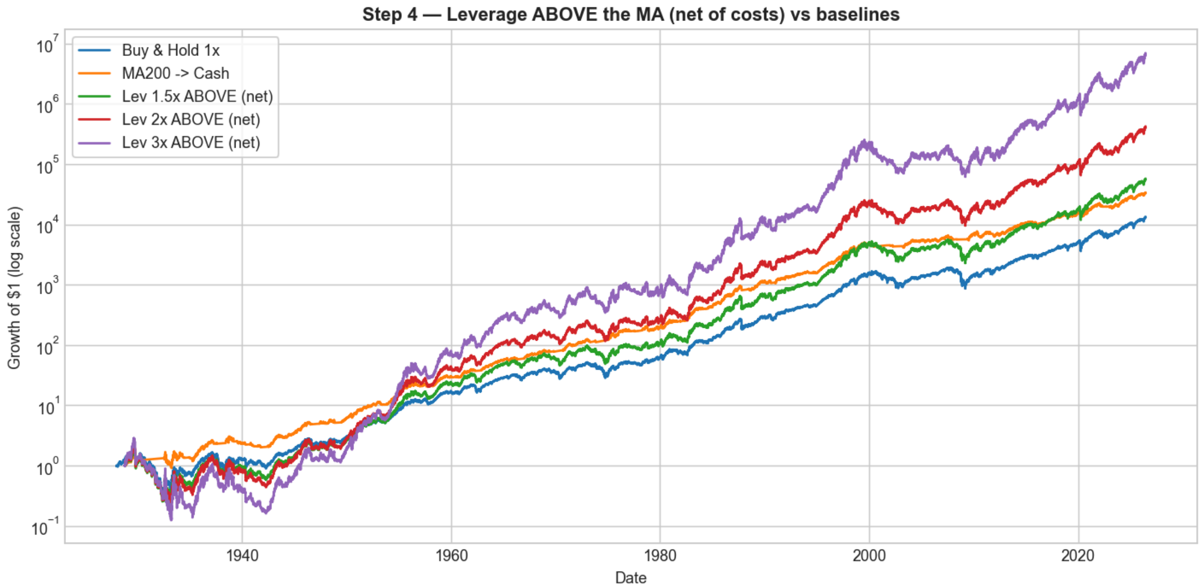
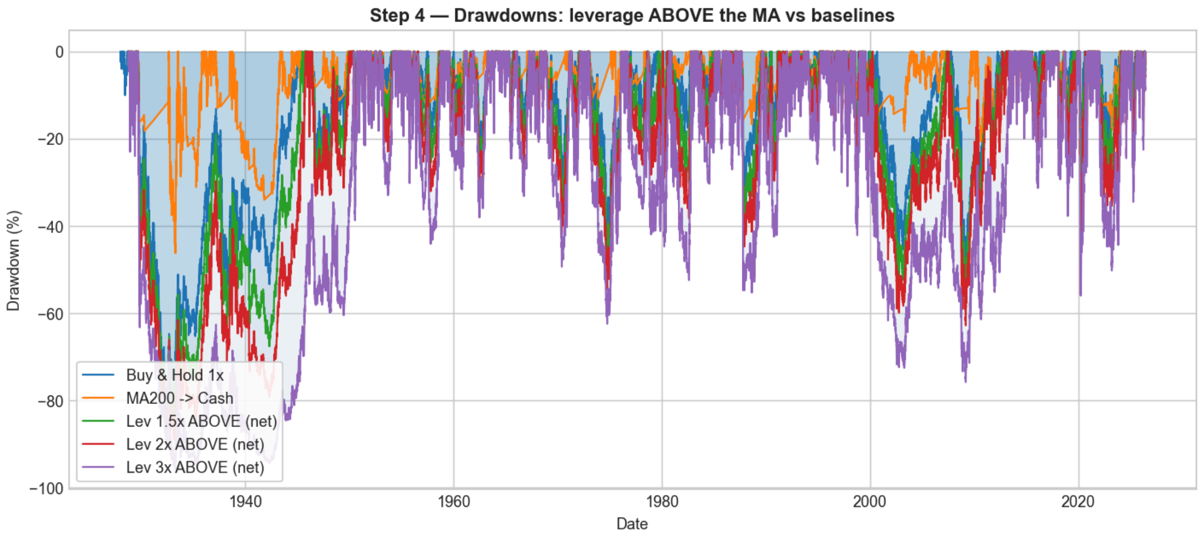
5. The switching strategy: leverage the uptrend

Use the 200-day MA to switch between leveraged and ordinary exposure: **hold L× leverage while the market is ABOVE the MA** — the calm, rising regime where the kind of one-directional gains seen in §3 actually happen — and drop back to plain 1× while it is below. Daily total return, 1928–2026, net of costs:

Strategy	Grew $1 to	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Buy & Hold 1×	$13,021	10.1%	18.9%	0.40	0.56	−83.9%	0.12
MA200 → Cash	$33,090	11.3%	12.6%	**0.60**	**0.84**	**−46.2%**	**0.24**
Lev 1.5× above MA	$55,471	11.9%	23.6%	0.43	0.60	−85.7%	0.14
Lev 2× above MA	$402,863	14.2%	28.9%	0.47	0.66	−89.2%	0.16
Lev 3× above MA	**$6,429,403**	**17.5%**	40.4%	0.50	0.71	−95.7%	0.18





Leveraging the uptrend gets **better** as leverage rises — $1 grows to $402,863 at 2× and $6.4M at 3× (versus $13,021 for buy & hold) — and it beats buy & hold on CAGR, Sharpe, Sortino and Calmar at every level. The trade-off: its maximum drawdown is deeper than buy & hold's (you are leveraged going *into* fast crashes), and it does not beat the plain move-to-cash rule on a risk-adjusted basis.

Educational research only — not investment advice. Reproduce with `python run_faber_leverage.py`*; figures are* *in `charts/`, numbers in `results/`.*